

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTAIES305 - DIGITAL LOGIC & SIGNAL PROCESSING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the primary purpose of a Digital Signal Processor (DSP)?
 - a) To convert analog signals to digital signals.
 - b) To perform mathematical operations on digital data streams.
 - c) To store large amounts of data.
 - d) To display information on a screen.
- 2 Which of the following is a key advantage of digital signal processing over analog signal processing?
 - a) Higher speed of operation
 - b) Lower cost
 - c) Greater flexibility and programmability
 - d) Simpler hardware implementation
- 3 When the frequency band is selected we can specify the sampling rate and the characteristics of the pre filter, which is also called as _____ filter.
 - a) Analog filter
 - b) Anti aliasing filter
 - c) Analog & Anti aliasing filter
 - d) None of the mentioned
- 4 The Fourier Transform of a unit impulse function ($\delta(n)$) is:
 - (a) A unit impulse function ($\delta(\omega)$).
 - (b) A constant value (1).
 - (c) A complex exponential function ($e^{-j\omega n}$).
 - (d) A unit step function ($u(\omega)$).
- 5 The area under the power spectral density (PSD) curve represents:
 - (a) The mean of the random process
 - (b) The variance of the random process
 - (c) The total power of the random process
 - (d) The correlation of the random process

- 6 The cross-spectral density of two random processes $X(t)$ and $Y(t)$ is defined as:
- The Fourier transform of the cross-correlation function $R_{xy}(\tau)$
 - The Fourier transform of the autocorrelation function $R_{xx}(\tau)$
 - The cross-correlation function $R_{xy}(\tau)$ itself
 - The autocorrelation function $R_{xx}(\tau)$
- 7 What is the decimal equivalent of the binary number 1101?
- 11
 - 12
 - 13
 - 14
- 8 What is the hexadecimal equivalent of the decimal number 27?
- 1A
 - 1B
 - 1C
 - 1D
- 9 What is the Boolean expression for the NAND gate?
- $Y = A.B$
 - $Y = A+B$
 - $Y = (A.B)'$
 - $Y = (A+B)'$
- 10 Which of the following is an example of a continuous-time signal?
- A digital audio file
 - A voltage signal from a sensor
 - A sequence of numbers representing stock prices
 - A video file
- 11 What is the primary difference between analog and digital signals?
- Analog signals are continuous, while digital signals are discrete
 - Digital signals are continuous, while analog signals are discrete
 - Analog and digital signals are the same
 - Both are continuous
- 12 What is the set of all values of z for which $X(z)$ attains a finite value?
- Radius of convergence
 - Radius of divergence
 - Feasible solution
 - None of the mentioned

Q2. Solve the following.

- Explain Digital Logic Circuit in details.
- Perform arithmetic operation
 $(101100101)_2 + (010011010)_2 = (11111111)_2$
 $(EC5)_{16} - (44)_{10}$

Q3. Solve the following.

- Define the term Signal. Explain Elementary Signals used for testing with its diagram
- Explain the difference between Continuous time and Discrete time signal

Q4. Solve Any Two of the following.

- Explain correlation and correlogram concept in brief

- B) Explain energy spectral density with its properties 6
- C) What do you mean by 1's complement and 2's complements of binary numbers? 6
Explain with suitable example

Q5. Solve Any Two of the following.

- A) Find Z transform of the following and indicate the ROC. i) $x(n) = \{1, 2, 3, 4\}$ 6
ii) $x(n) = a^n u(n), a > 1$
- B) Write a short note on sampling Process 6
- C) Solve any two examples of linear and circular 6

Q6. Solve Any Two of the following.

- A) What is mean by FFT(Fast Fourier Transform) Explain in brief. 6
- B) Explain the concept of convolution with its types 6
Draw and explain gate 6
i) NAND
ii) NOR
iii) EX-OR

*** End ***